

Maintenance Wizard Report

Diagnosis

Severe bearing degradation/failure (likely spalling, brinelling, or cage failure) within the CCM Segment Bearing, leading to increased friction, heat generation, and dynamic imbalance. This condition is indicative of terminal phase bearing life.

Root Cause

The most probable root causes are inadequate or contaminated lubrication, which leads to surface fatigue and wear. Other possibilities include misalignment, overloading, foreign particle ingress, or an inherent manufacturing defect exacerbated by operational stresses. Further investigation post-removal is required to pinpoint the exact cause.

Risk Level

Critical. High potential for catastrophic failure, significant damage to associated equipment (e.g., segment rollers, segment frame), extensive unplanned downtime, and safety hazards.

Failure Probability

Very High and rapidly increasing. The observed symptoms strongly suggest that the bearing is in the final stage of its operational life.

Remaining Useful Life

Extremely short (estimated in hours to a few days). Continued operation is highly unpredictable and carries a severe risk of sudden, complete failure.

Catastrophic Risk

High. Complete bearing seizure could lead to: structural damage to the casting segment, strand breakout, extensive unplanned production downtime (potentially weeks), secondary damage to adjacent mechanical components, and significant safety risks from sudden machine stoppage or flying debris.

Immediate Actions

[Implement continuous, close monitoring of vibration levels, bearing temperature, and abnormal noise (e.g., using thermal imaging, stethoscopes, online vibration analysis).', 'Notify production management and prepare for potential emergency shutdown and segment change-out.', 'Conduct a visual inspection for external signs of lubricant leakage, seal damage, or foreign material near the bearing.', 'Verify proper function of the lubrication system to the affected bearing (flow, pressure, lubricant quality if accessible).', 'Gather and review all available historical condition monitoring data (vibration trends, temperature trends, oil analysis if applicable).', 'Prepare necessary spares (CCM segment bearing, seals, fasteners) and tools for imminent replacement.']

Short Term Actions

['Schedule an immediate shutdown for bearing replacement at the earliest safe opportunity, prioritizing this over other maintenance tasks.', 'Safely remove the affected CCM segment from the caster.', 'Disassemble the segment to replace the faulty bearing and associated seals.', 'Thoroughly inspect the bearing housing, shaft, and adjacent rollers/components for any secondary damage.', 'Verify alignment of the new bearing and segment components during reassembly.', 'Ensure proper lubrication application (type, quantity) for the new bearing.', 'Perform functional checks, dynamic balancing (if applicable to segment rollers), and a controlled run-in period post-replacement.']

Long Term Actions

['Conduct a detailed root cause analysis on the failed bearing and collect samples (e.g., wear debris, lubricant) for metallurgical and tribological analysis.', 'Review and optimize the lubrication regime for CCM segment bearings (lubricant type, frequency, volume, delivery system reliability).', 'Enhance the predictive maintenance program for CCM segments, focusing on continuous vibration monitoring, periodic thermal imaging, and potentially integrated oil analysis.', 'Evaluate the design, material, or supplier of current bearings if failures are recurrent, considering upgraded alternatives (e.g., sealed bearings, higher load capacity).', 'Improve training for maintenance personnel on proper bearing installation, lubrication practices, and fault diagnosis techniques.', 'Implement stricter control over cleanliness and contamination prevention during maintenance activities.']

Spare Procurement

['Ensure immediate availability of at least one complete CCM segment bearing and associated seal kit for emergency replacement.', 'Establish robust re-ordering points and lead times to maintain critical spare inventory levels for CCM segment bearings.', 'Consider the strategic procurement of a pre-assembled spare CCM segment to minimize future segment change-out times during unplanned outages.']

Executive Summary

A critical CCM segment bearing is exhibiting severe and escalating symptoms of high vibration, increasing temperature, and abnormal noise, indicating imminent failure. This presents a critical risk of catastrophic equipment damage, extensive unplanned downtime, and significant safety hazards. Immediate action is required to monitor the condition, prepare for an emergency shutdown, and replace the affected bearing. A comprehensive root cause analysis and implementation of long-term preventive measures are essential to mitigate recurrence and ensure plant reliability.

Maintenance Recommendation

It is strongly recommended to immediately escalate this issue, prepare for an emergency shutdown, and prioritize the replacement of the affected CCM segment bearing. This proactive intervention is crucial to prevent catastrophic failure, minimize secondary damage, and reduce the duration of unplanned downtime. Following replacement, a thorough root cause analysis must be conducted to identify and address underlying issues related to lubrication, alignment, or contamination, complemented by an enhanced predictive maintenance strategy.

Conclusion

The CCM segment bearing is in an advanced state of failure, posing an extreme risk to operations. Continued running is highly inadvisable and will almost certainly lead to a catastrophic event.

Urgent and decisive action, including immediate preparation for and execution of bearing replacement, is imperative to safeguard plant assets, ensure personnel safety, and maintain production continuity.